

Statistical Analysis of Attachment Time in an Ingestible Drug Delivery Robot

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Table of Contents

1. Introduction.....	3
2. Methods This section describes the experimental procedure and statistical analyses used.....	3
2.1 Experimental Procedure.....	3
2.2 Statistical Analysis.....	3
3. Results This section provides the findings derived from the statistical analysis.....	3
3.1 Descriptive Statistics.....	3
3.2 Assumption Checks.....	4
3.3 Two-Way ANOVA.....	4
3.4 Interaction Plot.....	5
3.5 Post-Hoc Analysis.....	5
4. Discussion.....	6
This section interprets the statistical findings, justifies analytical choices made, and considers approach limitations.....	6
4.1 Addressing the Research Questions.....	6
4.2 Two-Way ANOVA: Justification, Advantages and Disadvantages.....	6
4.3 Choice of Tests and Comparisons with Alternatives.....	6
4.4 Limitations.....	7
References.....	8
Appendices.....	9
Appendix A: Normal Probability Plot of Residuals (Ryan-Joiner Test).....	9
Appendix B: Levene’s Test for Equal Variances-Interval Plot.....	9

1. Introduction

This report analyses how orifice diameter and vacuum volume affect an ingestible drug delivery robot's attachment time inside the small intestine, and their interaction.

2. Methods

This section describes the experimental procedure and statistical analyses used.

2.1 Experimental Procedure

The experiment used a 3×2 full factorial design, testing three orifice diameters (4 mm, 5 mm and 6 mm) against two vacuum volumes (0.4 mm³ and 0.6 mm³), giving six conditions each repeated three times, with a total of 18 observations. The outcome variable was total attachment time in hours. Descriptive statistics, including the mean, standard deviation, minimum and maximum, were first calculated for each condition to summarize the data before any formal testing.

2.2 Statistical Analysis

A two-way analysis of variance (ANOVA) was selected as the primary statistical method. A significance level of $\alpha = 0.05$ was applied throughout. All analyses were carried out using Minitab Statistical Software (Version 21; Minitab LLC, 2024).

Three parametric assumptions were checked on the model residuals. First, normality of residuals was assessed using the Ryan-Joiner test (Ryan and Joiner, 1976). Second, homogeneity of variance was assessed using Levene's test (Levene, 1960). Third, independence of observations was satisfied by design. Since the main effect of orifice diameter proved significant, a Tukey's Honestly Significant Difference (HSD) post-hoc test was carried out to identify which specific diameters differed from one another (Tukey, 1949).

3. Results

This section provides the findings derived from the statistical analysis.

3.1 Descriptive Statistics

As a first step, descriptive statistics were calculated for each condition in Minitab and are shown in Table 1. The 6 mm diameter produced the longest attachment times across both vacuum volume conditions (38.23 hours and 39.90 hours), while the 4 mm and 5 mm conditions were considerably lower and similar to one another. The larger vacuum volume added attachment time consistently across all diameters. The standard deviations for the 6 mm conditions (1.75 and 2.70) were notably larger than for the other groups (0.50 to 0.81), which suggests greater variability at the largest diameter. These patterns are explored through formal testing in the sections that follow.

Table 1. Descriptive statistics for attachment time (hours) by condition, produced in Minitab (Version 21).

Orifice Diameter (mm)	Vacuum Volume (mm ³)	N	Mean (hours)	± SD (hours)	Min (hours)	Max (hours)
4	0.4	3	27.9000	0.5000	27.4	28.4
4	0.6	3	30.2333	0.7638	29.4	30.9
5	0.4	3	29.7333	0.7638	28.9	30.4
5	0.6	3	31.3667	0.8083	30.9	32.3
6	0.4	3	38.2333	1.7474	36.3	39.7
6	0.6	3	39.9000	2.6963	36.8	41.7

3.2 Assumption Checks

Before carrying out the ANOVA, the three parametric assumptions were formally checked. The results are summarised in Table 2 and supported visually by the plots in Appendix A and Appendix B. The Ryan-Joiner test returned $RJ = 0.966$ with $p > 0.100$, indicating no significant departure from normality in the residuals. Levene's test returned $F = 0.55$ with $p = 0.734$, confirming that variances were equal across all six groups. Minitab's multiple comparisons method produced a consistent result of $p = 0.297$. Although the 6 mm group showed wider spread descriptively, neither test detected a significant violation. This is most likely because three observations per group gives these tests limited statistical power to detect violations. As all assumptions were met, the two-way ANOVA was appropriate to proceed with, and the results are presented below.

Table 2. Summary of parametric assumption checks prior to two-way ANOVA, conducted in Minitab (Version 21).

Assumption	Test Used	Test Statistic	P-Value	Decision
Normality of residuals	Ryan-Joiner test	$RJ = 0.966$	> 0.100	Met. Proceed with ANOVA
Homogeneity of variance	Levene's test	$F = 0.55$	0.734	Met. Proceed with ANOVA
Independence of observations	Satisfied by design	Not applicable	Not applicable	Met. No test required

3.3 Two-Way ANOVA

The two-way ANOVA results are presented in Table 3, along with partial eta-squared (η^2) as a measure of effect size. Orifice diameter had a highly significant and very large effect on attachment time ($F(2, 12) = 84.59$, $p < 0.001$, $\eta^2 = 0.934$). To put this plainly, if orifice diameter had no true effect, the probability of observing an F-statistic as large as 84.59 purely by chance would be less than 0.001. Vacuum volume was also significant ($F(1, 12) = 7.68$, $p = 0.017$, $\eta^2 = 0.390$), representing a moderate to large effect. The interaction between the two factors showed no statistically significant effect ($F(2, 12) = 0.11$, $p = 0.894$, $\eta^2 = 0.018$), which means no interaction between diameter and vacuum volume was detected. The overall model explained 93.65% of the total variation in attachment time ($R^2 = 93.65\%$). Observation 16 was flagged by Minitab as an unusual observation (standardized residual = -2.64), corresponding to a recorded time of 36.8 hours in the 6 mm / 0.6 mm³ condition. This is consistent with the higher variability seen at the 6 mm diameter and is explored further in the post-hoc analysis below.

Table 3. Two-way ANOVA results including partial eta-squared effect sizes, generated in Minitab (Version 21). *DF* = degrees of freedom; *Adj SS* = adjusted sum of squares; *Adj MS* = adjusted mean square; η^2 = partial eta-squared.

Source	DF	Adj SS	Adj MS	F-Value	P-Value	Partial η^2
Diameter	2	349.468	174.734	84.59	< 0.001	0.934
Vacuum Volume	1	15.867	15.867	7.68	0.017	0.390
Diameter $\times$ Vacuum Volume	2	0.468	0.234	0.11	0.894	0.018
Error	12	24.787	2.066	—	—	—
Total	17	390.589	—	—	—	—

3.4 Interaction Plot

To confirm the non-significant interaction obtained from ANOVA ($p = 0.894$), an interaction plot was produced in Minitab. As shown in figure 1, the lines for each vacuum volume condition run approximately parallel across all three orifice diameters, which indicates no interaction between the two factors on attachment time ($p = 0.894$). The sharp increase in attachment time moving from the 5 mm to the 6 mm diameter occurred consistently across both vacuum volume levels, providing visual confirmation that the two factors act independently of one another.

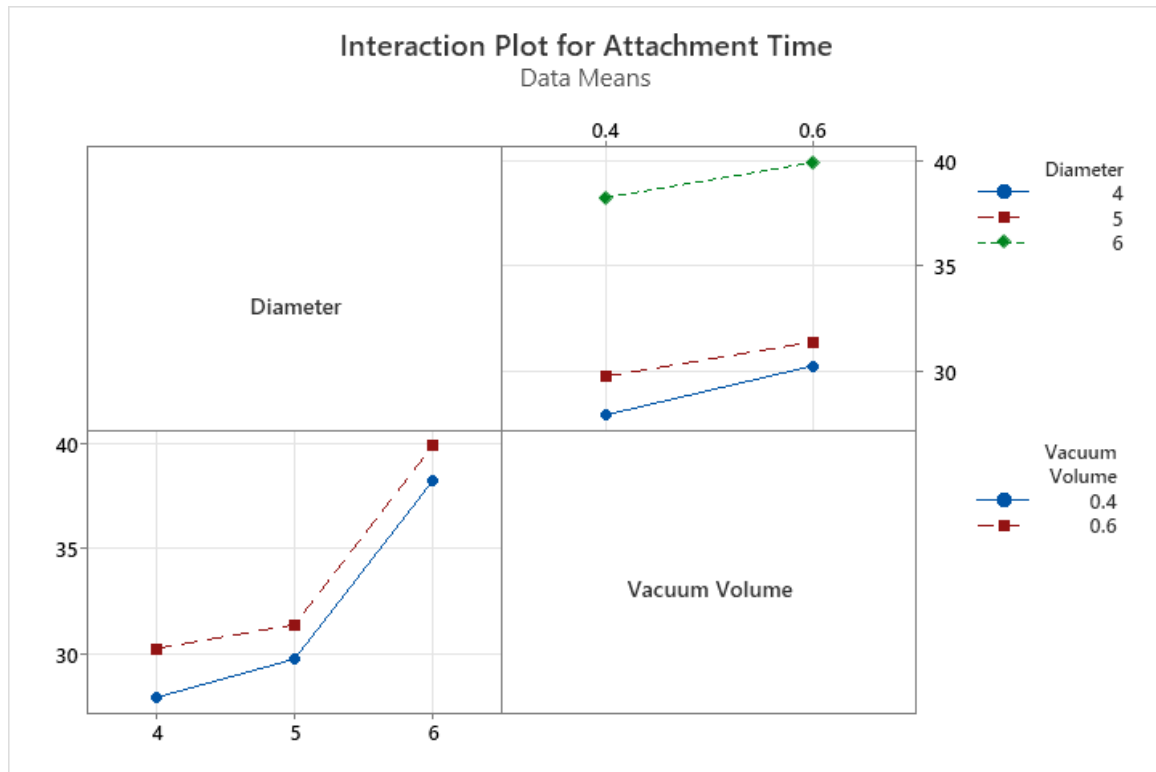


Figure 1. Interaction plot for mean attachment time by orifice diameter and vacuum volume (Minitab, Version 21). The approximately parallel lines show that increased diameter and larger vacuum volume each improve attachment time independently, with no combined effect between them ($p = 0.894$).

3.5 Post-Hoc Analysis

Since the main effect of orifice diameter was significant across three levels, Tukey's HSD post-hoc test was carried out to determine which specific diameters differed from one another. The results are shown in Table 4. The 6 mm diameter (overall mean = 39.07 hours, group A) produced significantly longer attachment times than both the 5 mm (30.55 hours) and 4 mm (29.07 hours) diameters, both of which fell into group B. The 4 mm and 5 mm diameters were not significantly different from one another. These results are discussed and interpreted in the section below.

Table 4. Tukey's HSD post-hoc grouping for orifice diameter. Groups not sharing a letter are significantly different from one another (Minitab, Version 21).

Orifice Diameter	N	Overall Mean (hours)	Tukey Group	Conclusion
6 mm	6	39.07	A	Significantly longer than 4 mm and 5 mm
5 mm	6	30.55	B	Not significantly different from 4 mm
4 mm	6	29.07	B	Not significantly different from 5

				mm
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4. Discussion

This section interprets the statistical findings, justifies analytical choices made, and considers approach limitations.

4.1 Addressing the Research Questions

Building on the results above, the three research questions are directly answered. First, the 6 mm orifice diameter produces the longest mean attachment time (39.07 hours), significantly outperforming both smaller diameters. Second, the larger vacuum volume of 0.6 mm³ also produces longer attachment times ($p = 0.017$). Third, no significant interaction was detected ($p = 0.894$), meaning the two factors act independently, and the optimal configuration is a 6 mm orifice with 0.6 mm³ vacuum volume. Using Cohen's benchmarks, where partial η^2 above 0.14 is large (Cohen, 1988), the diameter effect ($\eta^2 = 0.934$) is exceptionally large and vacuum volume ($\eta^2 = 0.390$) is large (Richardson, 2011). The interaction effect was very small ($\eta^2 = 0.018$), and the study had limited power to detect an effect of this size. The non-significant interaction should therefore be read as no interaction was detected, not as proof that none exists. These findings are consistent with the broader principle that mechanical parameters critically determine retention in ingestible devices for targeted drug delivery (Montgomery, 2017).

4.2 Two-Way ANOVA: Justification, Advantages and Disadvantages

A two-way ANOVA was the most appropriate method because it tests both main effects and their interaction within a single model, avoiding the inflated false positive risk of running separate one-way ANOVAs (Kim, 2014; Field, 2018). Both factors were treated as fixed effects and the interaction term was included from the outset, since removing it prematurely would bias the main effect estimates. The R^2 of 93.65% confirms the model captured the vast majority of variation. Assumptions were checked on the model residuals because ANOVA assumptions apply to residual errors, not the raw data. Key advantages include detecting interactions and efficiency with factorial designs; disadvantages include sensitivity to assumption violations in small samples and inability to handle missing data (Glass, Peckham and Sanders, 1972). Had assumptions been violated, the Scheirer-Ray-Hare test would have been considered, though it reduces statistical power and has no widely accepted effect size measure. More modern alternatives such as permutation tests and bootstrapping offer greater robustness while preserving effect size estimation.

4.3 Choice of Tests and Comparisons with Alternatives

The Ryan-Joiner test was used for normality because it is the test available within the Minitab normality test menu for residuals (Ryan and Joiner, 1976). Shapiro-Wilk would have been preferable as it is more widely cited in biomedical research with slightly greater statistical power at small sample sizes (Razali and Wah, 2011), but Ryan-Joiner is a well-established alternative. Anderson-Darling was not selected as it weights distributional tails more heavily and suits larger datasets. With only 18 residuals, any normality test has limited power, so a non-significant result should not be taken as proof of normality (Field, 2018). The independence of observations assumption was satisfied by design since each trial used a fresh independent sample with no repeated measures. For variance equality, Levene's test was preferred over Bartlett's because Bartlett's is highly sensitive to departures from normality (Levene, 1960). For post-hoc comparisons, Tukey's HSD was chosen over the Bonferroni correction (Bonferroni, 1936) because Bonferroni becomes

increasingly conservative as comparisons grow. Tukey's HSD is purpose-built for pairwise comparisons after ANOVA and maintains better statistical power (Field, 2018).

4.4 Limitations

The small sample size (three per condition) reduced statistical power, and without a formal power analysis it is unclear whether the sample was sufficient. Although diameter and vacuum volume showed large effects, the small interaction ($f = 0.14$) was likely underpowered (Cohen, 1988), so non-significance does not prove absence. Effect sizes may be inflated (Richardson, 2011), and unequal group variability undetected by Levene's test, along with controlled laboratory conditions, may limit how well the findings reflect performance in a living gastrointestinal tract.

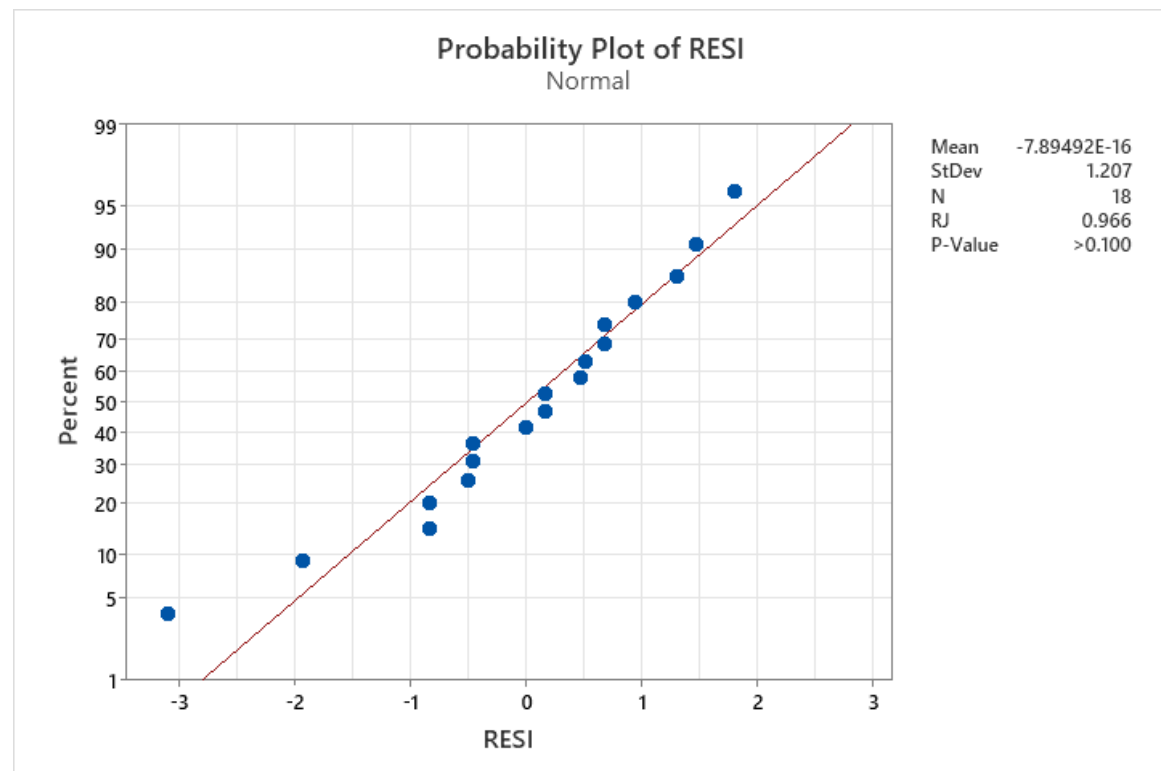
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Appendices

Appendix A: Normal Probability Plot of Residuals (Ryan-Joiner Test)

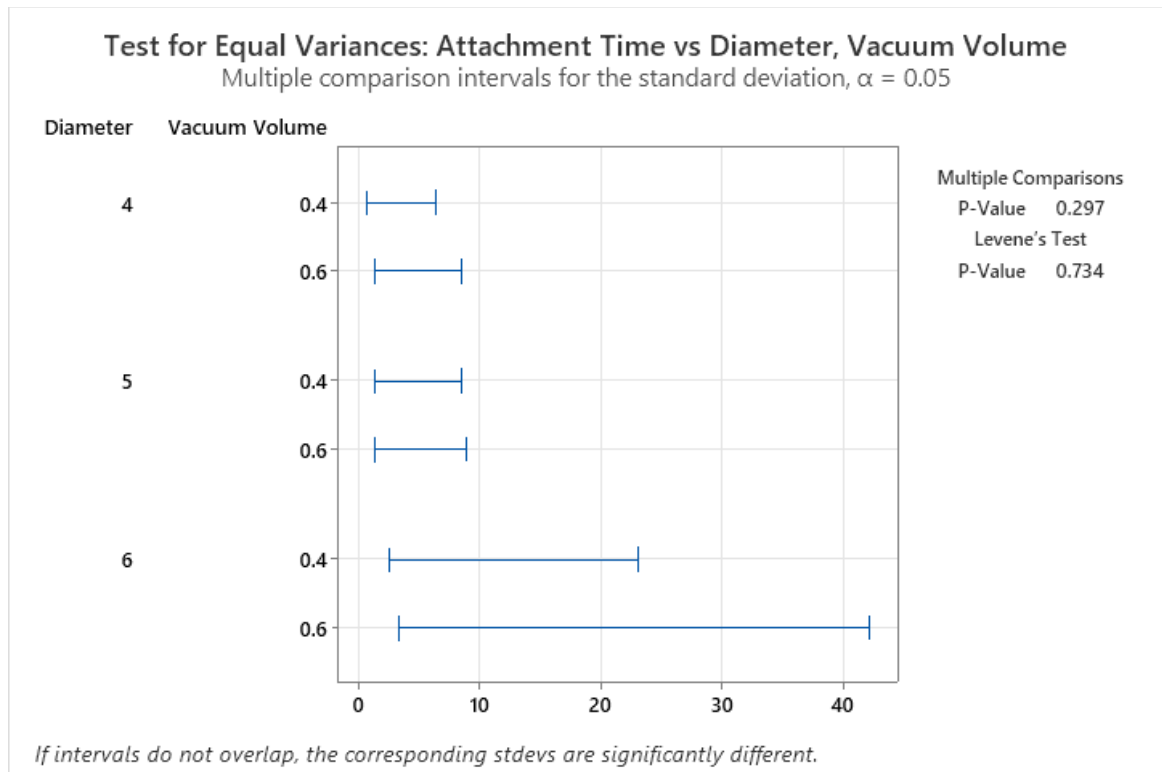
The normal probability plot below was produced in Minitab by running the Ryan-Joiner normality test on the saved ANOVA residuals (Stat → Basic Statistics → Normality Test, selecting the RESI column). The data points fall closely along the reference line throughout the range, which visually confirms the formal test result ($RJ = 0.966$, $p > 0.100$) and supports the conclusion that the residuals do not depart significantly from a normal distribution.



Appendix Figure A. Probability plot of ANOVA residuals (Minitab, Version 21). Points close to the reference line support the assumption of normality. $RJ = 0.966$, $p > 0.100$.

Appendix B: Levene's Test for Equal Variances-Interval Plot

The interval plot below shows the 95% confidence intervals for the standard deviation of each condition, produced in Minitab. The intervals overlap across all six conditions, which is consistent with the Levene's test result ($F = 0.55$, $p = 0.734$) and confirms that variances are not significantly different across groups. The notably wider intervals for the 6 mm conditions are consistent with the larger standard deviations reported in Table 1 and the unusual observation flagged in Section 3.3.



Appendix Figure B. Interval plot of standard deviations by condition (Minitab, Version 21). Overlapping intervals confirm homogeneity of variance across all six groups (Levene's $p = 0.734$).